

2023 AMC 10B Problem 9 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 9

Problem:

The numbers 16 and 25 are a pair of consecutive positive perfect squares whose difference is 9. How many pairs of consecutive positive perfect squares have a difference of less than or equal to 2023?

Answer Choices:

- A. 674
- B. 1011
- C. 1010
- D. 2019
- E. 2017

Solution:

If the consecutive positive perfect squares are n^2 and $(n + 1)^2$, then their difference is $(n^2 + 2n + 1) - n^2 = 2n + 1$. Solving $2n + 1 \leq 2023$ gives $n \leq 1011$, so there are (B) 1011 pairs of consecutive positive square pairs whose difference is less than or equal to 2023, namely $\{1^2, 2^2\}$, $\{2^2, 3^2\}$, $\{3^2, 4^2\}$, $\dots$, $\{1011^2, 1012^2\}$

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